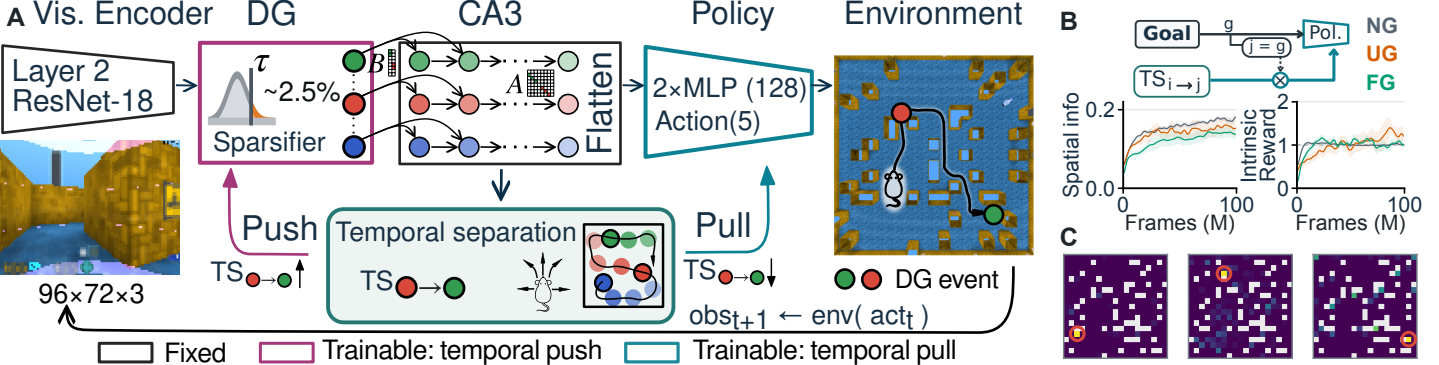
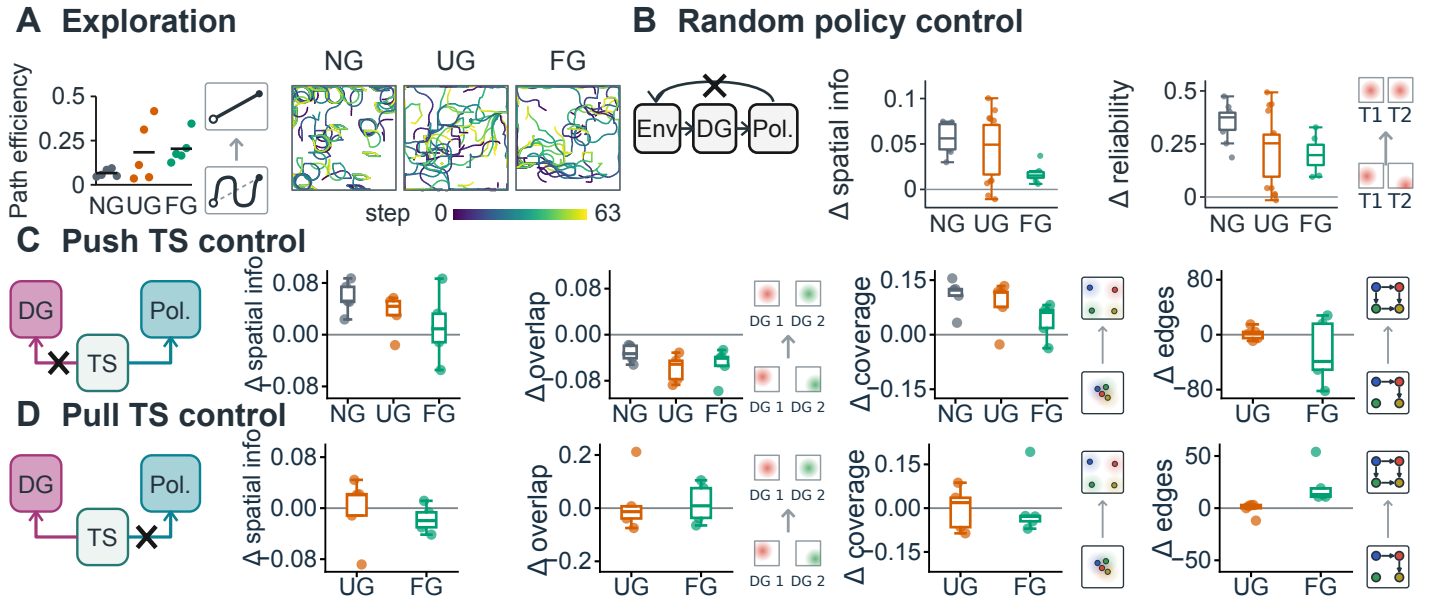

A Vis. Encoder
DG
CA3
Policy
Environment
Layer 2
ResNet-18
96×72×3
τ
~2.5%
Sparsifier
B
A
Flatten
2×MLP (128)
Action(5)
Push
TS
Temporal separation
Pull
DG event
obs$_{t+1}$ ← env(act$_t$)
Fixed
Trainable: temporal push
Trainable: temporal pull
B
Goal
g
j = g
Pol.
TS$_{i→j}$
NG
UG
FG
Spatial info
0.2
0.0
0
100
Frames (M)
Intrinsic Reward
2
1
0
100
Frames (M)
C


A Exploration
Path efficiency
0.5
0.25
0
NG UG FG
NG
UG
FG
step 0 63
B Random policy control
Env→DG→Pol.
Δ spatial info
0.1
0.05
0
NG UG FG
Δ reliability
0.5
0.25
0
NG UG FG
T1 T2
T1 T2
C Push TS control
DG
Pol.
TS
Δ spatial info
0.08
0.00
−0.08
NG UG FG
Δ overlap
0.08
0.00
−0.08
NG UG FG
DG 1 DG 2
DG 1 DG 2
Δ coverage
0.15
0.00
−0.15
NG UG FG
Δ edges
80
0
−80
UG FG
D Pull TS control
DG
Pol.
TS
Δ spatial info
0.08
0.00
−0.08
UG FG
Δ overlap
0.2
0.0
−0.2
UG FG
DG 1 DG 2
DG 1 DG 2
Δ coverage
0.15
0.00
−0.15
UG FG
Δ edges
50
0
−50
UG FG